6347 - A High-Performance Analysis Framework for JWST Exoplanet Observations

Cycle: 3, Proposal Category: AR

INVESTIGATORS

Name	*Institution*
Dr. Matthew Conor Nixon (PI)	**University of Maryland**
Dr. Jake Taylor (CoI) (ESA Member)	University of Oxford
Mr. Jegug Ih (CoI)	University of Maryland
Dr. Luis Welbanks (CoI)	Arizona State University
Dr. Megan Mansfield (CoI)	University of Arizona
Dr. Michael Line (CoI)	Arizona State University
Peter Smith (CoI)	Arizona State University
Dr. Vivien Parmentier (CoI) (ESA Member)	Universite Cote d Azur
Dr. Peter McGill (CoI)	Lawrence Livermore National Laboratory
Prof. Eliza M.-R. Kempton (CoI)	University of Maryland

OBSERVATIONS

ABSTRACT

The first JWST data sets are providing unprecedented insight into a range of physical and chemical processes that take place in exoplanet atmospheres. These data mark the beginning of a revolution in our understanding of the phenomena that shape planetary formation and evolution. One of the key algorithms used to infer atmospheric properties from an observed spectrum is atmospheric retrieval, which can obtain constraints on key atmospheric properties such as chemical abundances and temperature profiles. While this approach has been successful in the past, a number of key improvements are urgently required to maximise the scientific output from JWST observations of exoplanets and prevent biased inferences. Atmospheric retrieval is computationally intensive, and requires heavily simplified atmospheric models in order to be feasible. This prevents us from learning about more complex atmospheric processes that are left out of the models, as well as creating the potential for biased inferences due to missing physics.

The overall aim of the proposed research is to create a robust, high-performance retrieval framework that surmounts these challenges. This can be broken down into two key objectives: (1) improving the computational efficiency of the retrieval algorithm to enable the use of more complex forward models, and (2) integrating 3D models of exoplanet atmospheres into the retrieval framework, since 1D models are known to introduce biases. These developments will significantly advance our ability to characterise exoplanet atmospheres using JWST observations, setting the stage for a revolution in our understanding of other worlds.

OBSERVING DESCRIPTION

N/A (this is a theory proposal).